Study Guide

Expect a **subset** of questions drawn from the following topics.

Linear Regression Concepts

- A conceptual question describing and interpreting the effect of mixing continuous and categorical variables in linear regression, as discussed in Chapter 3 of ISLP.
- A question requiring you to list **at least five assumptions** associated with linear regression.
- A question on **heteroscedasticity**: what it is, why it is problematic in linear regression, and how it affects results and inference.
- A question in which you are given a **nonlinear relationship** and asked to linearize it by defining engineered features and writing the resulting linear regression model in the transformed space (e.g. polynomial regression).

Logistic Regression Theory

1. Given the equation for **categorical cross-entropy**, show that it reduces to **binary cross-entropy** when (C = 2) (number of classes). $-\sum_{i=1}^{n}\left[y_i \log(p_i) + (1 - y_i)\log(1 - p_i)\right]$

appendix1
2. Work through a **loss function calculation** (either BCE or CCE) for a small toy classification dataset.

appendix2
3. Starting from the logit formulation, derive an explicit expression for (p(X)):

$$\log\left(\frac{p(X)}{1 - p(X)}\right) = \beta_0 + \beta^T X \tag{1}$$

appendix3
4. Compute the derivative of the **sigmoid function**.
5. Label a sketch of the sigmoid function given different values of fitting parameters (e.g., match parameter values to the appropriate curve).
6. **Bonus:** Derive an analytical expression for the derivative of the BCE loss with respect to the parameters of a univariate logistic regression model.

Interpreting Logistic Regression Results

Be familiar with the following aspects of interpreting logistic regression output. Expect **conceptual multiple-answer** or **short-answer** questions.

1. **Estimated Coefficients (($\hat{\beta}$))**
   - Represent how each predictor influences the outcome on the **log-odds** scale
   - Define the fitted model and encode both the direction (increase/decrease) and strength of each effect
   - A one-unit increase in (X_j) changes the log-odds of (Y = 1) by ($\hat{\beta}_j$), holding other variables fixed

2. **Standard Errors**
    - Quantify the **uncertainty** in coefficient estimates (derived from the observed Fisher information)
    - Form the basis for confidence intervals and hypothesis tests
    - Large standard errors suggest unstable estimates due to multicollinearity, small sample size, or separation
3. **Wald Z-Statistics and P-values**
    - Test the null hypothesis: $H_0 : \beta_j = 0$
    - Measure how many standard errors the estimated coefficient is from zero
    - Small p-values indicate statistically significant evidence of a nonzero effect
4. **Odds Ratios and Confidence Intervals**
    - Obtained by exponentiating coefficients: $e^{\hat{\beta}_j}$
    - Provide a more intuitive, multiplicative interpretation of effects on the odds scale
    - An odds ratio (> 1) increases the odds of (Y = 1); an odds ratio (< 1) decreases the odds
5. **Confidence Intervals for Predicted Probabilities**
    - Quantify uncertainty in the predicted probability $\hat{p}(X)$ for specific predictor values
    - Distinguish point predictions from uncertainty-aware probability estimates
    - Narrow intervals indicate stable predictions; wide intervals often occur near separation or extrapolation

Appendix

appendix1

Example 1: Binary Cross-Entropy (BCE)

Setup

- Binary classification: $y \in \{0, 1\}$
- Loss for one observation:

$$\mathrm{BCE}(y, p) = -[y \log(p) + (1 - y)\log(1 - p)]$$

Toy dataset (3 observations)

Observation	True label y	Predicted prob $p = P(y = 1)$
1	1	0.9
2	0	0.3
3	1	0.6

Step-by-step loss calculation

Observation 1

$$\mathrm{Loss}_1 = -\log(0.9) = 0.105$$

Observation 2

$$\mathrm{Loss}_2 = -\log(1 - 0.3) = -\log(0.7) = 0.357$$

Observation 3

$$\mathrm{Loss}_3 = -\log(0.6) = 0.511$$

Total and average loss

$$\text{Total BCE} = 0.105 + 0.357 + 0.511 = 0.973$$

$$\text{Average BCE} = \frac{0.973}{3} = 0.324$$

Example 2: Categorical Cross-Entropy (CCE)

Setup

- 3 classes: $C = 3$
- One-hot encoded labels
- Loss for one observation:

$$\mathrm{CCE} = -\sum_{c=1}^{C} y_c \log(p_c)$$

Toy dataset (2 observations)

Observation 1

- True class: **Class 2**
- One-hot label:

$$y = [0, 1, 0]$$

- Predicted probabilities:

$$p = [0.2, 0.5, 0.3]$$

Loss:

$$\mathrm{Loss}_1 = -\log(0.5) = 0.693$$

Observation 2

- True class: **Class 1**
- One-hot label:

$$y = [1, 0, 0]$$

- Predicted probabilities:

$$p = [0.7, 0.2, 0.1]$$

Loss:

$$\mathrm{Loss}_2 = -\log(0.7) = 0.357$$

Total and average loss

$$\text{Total CCE} = 0.[illegible] + 0.357 = 1.050$$

appendix2

Given (logit formulation)

$$\log\left(\frac{p(x)}{1 - p(x)}\right) = \beta_0 + \beta^\top x$$

where:

- $p(x) = P(Y = 1 \mid X = x)$
- $\beta^\top x$ is the dot product of coefficients and predictors

Step 1: Exponentiate both sides

$$\frac{p(x)}{1 - p(x)} = e^{\beta_0 + \beta^\top x}$$

Step 2: Solve for $p(x)$

Multiply both sides by $1 - p(x)$:

$$p(x) = (1 - p(x))e^{\beta_0 + \beta^\top x}$$

Expand:

$$p(x) = e^{\beta_0 + \beta^\top x} - p(x)e^{\beta_0 + \beta^\top x}$$

Step 3: Collect $p(x)$ terms

$$p(x)\left(1 + e^{\beta_0 + \beta^\top x}\right) = e^{\beta_0 + \beta^\top x}$$

Step 4: Final expression

$$\boxed{p(x) = \frac{e^{\beta_0 + \beta^\top x}}{1 + e^{\beta_0 + \beta^\top x}}}$$

Equivalent (often preferred) form

$$\boxed{p(x) = \frac{1}{1 + e^{-(\beta_0 + \beta^\top x)}}}$$

appendix3

$$\frac{d}{dz}\sigma(z)$$

2️⃣ Step 1: Rewrite for easier differentiation

$$\sigma(z) = (1 + e^{-z})^{-1}$$

3️⃣ Step 2: Use the chain rule

$$\frac{d}{dz}(1 + e^{-z})^{-1} = -(1 + e^{-z})^{-2} \cdot \frac{d}{dz}(1 + e^{-z})$$

Derivative of $1 + e^{-z}$ is:

$$\frac{d}{dz}(1 + e^{-z}) = -e^{-z}$$

So we get:

$$\frac{d}{dz}\sigma(z) = -(1 + e^{-z})^{-2} \cdot (-e^{-z}) = \frac{e^{-z}}{(1 + e^{-z})^2}$$

4️⃣ Step 3: Express in terms of $\sigma(z)$

Notice:

$$\sigma(z) = \frac{1}{1 + e^{-z}} \implies 1 - \sigma(z) = \frac{e^{-z}}{1 + e^{-z}}$$

Then:

$$\frac{e^{-z}}{(1 + e^{-z})^2} = \sigma(z) \cdot (1 - \sigma(z))$$

✅ 5️⃣ Final derivative

$$\boxed{\frac{d}{dz}\sigma(z) = \sigma(z)\,(1 - \sigma(z))}$$